

Amp Meter Bro

A Smart Power Metering Device Prototype

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Motivation

The continued growth of modern computing systems has driven a rising demand for efficient, intelligent, and reliable power electronics. This portable power monitoring device is designed to provide hands-on experience with the fundamental principles behind modern smart metering systems. By integrating embedded hardware, sensing circuitry, and real-time data visualization, the project demonstrates how voltage, current, and power consumption can be monitored in a compact platform.



Fig. 1 Schneider Electric Power Logic PM3000 Power Meter
PowerLogic™ PM3000 Power Meters | Schneider Electric USA

Future Work

- Deploy the real-time data visualization website as digital dashboard
- Add data tracking functionalities to digital dashboard
- Scale up current hardware design to handle two-phase/three-phase power system
- Enable not only over-current/power detection, but at the same time allowing under-current detection
- Customize PCB with integrated DC/DC converter and microcontroller.

Amp Meter Bro Power Monitor

Voltage: 7.24 V
Current: 18.90 mA
Power: 136.84 mW
Status: **NORMAL**

Fig.4 The Digital Dashboard (Undeployed)

System Level Overview

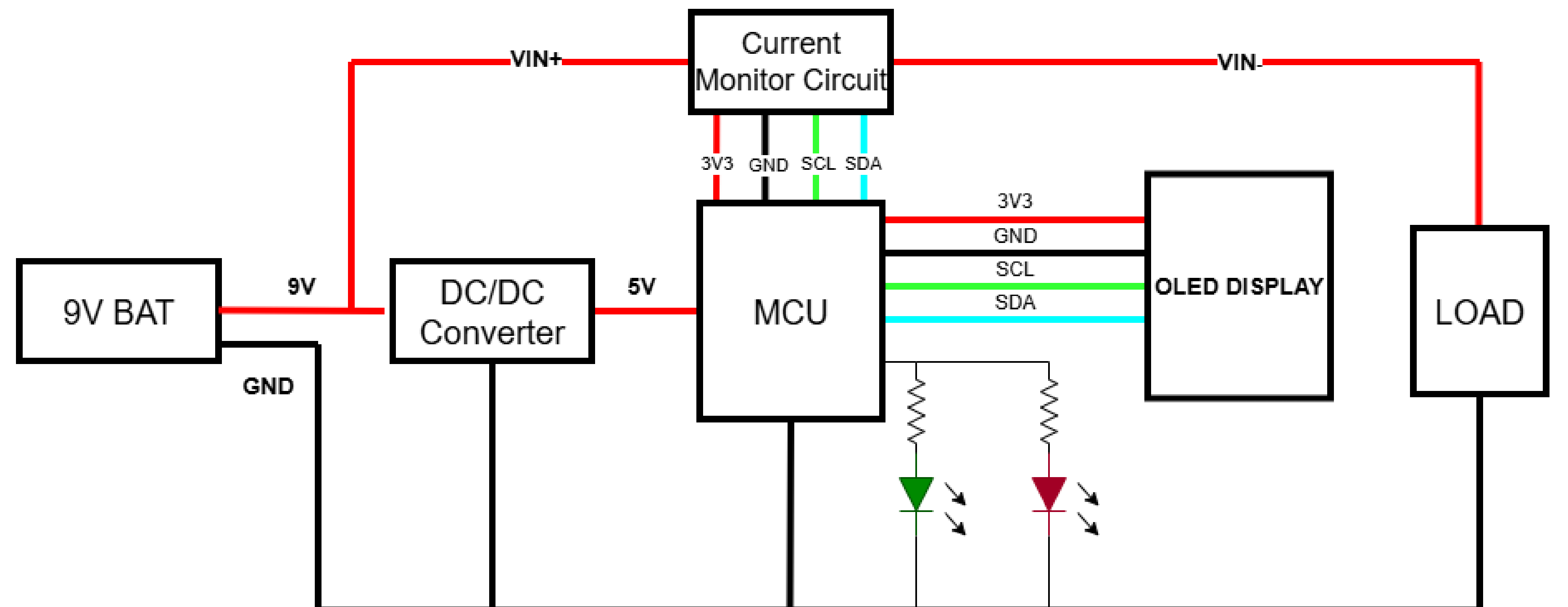


Fig.2 System Level Hardware Breakdown

Current Monitor Circuit Design

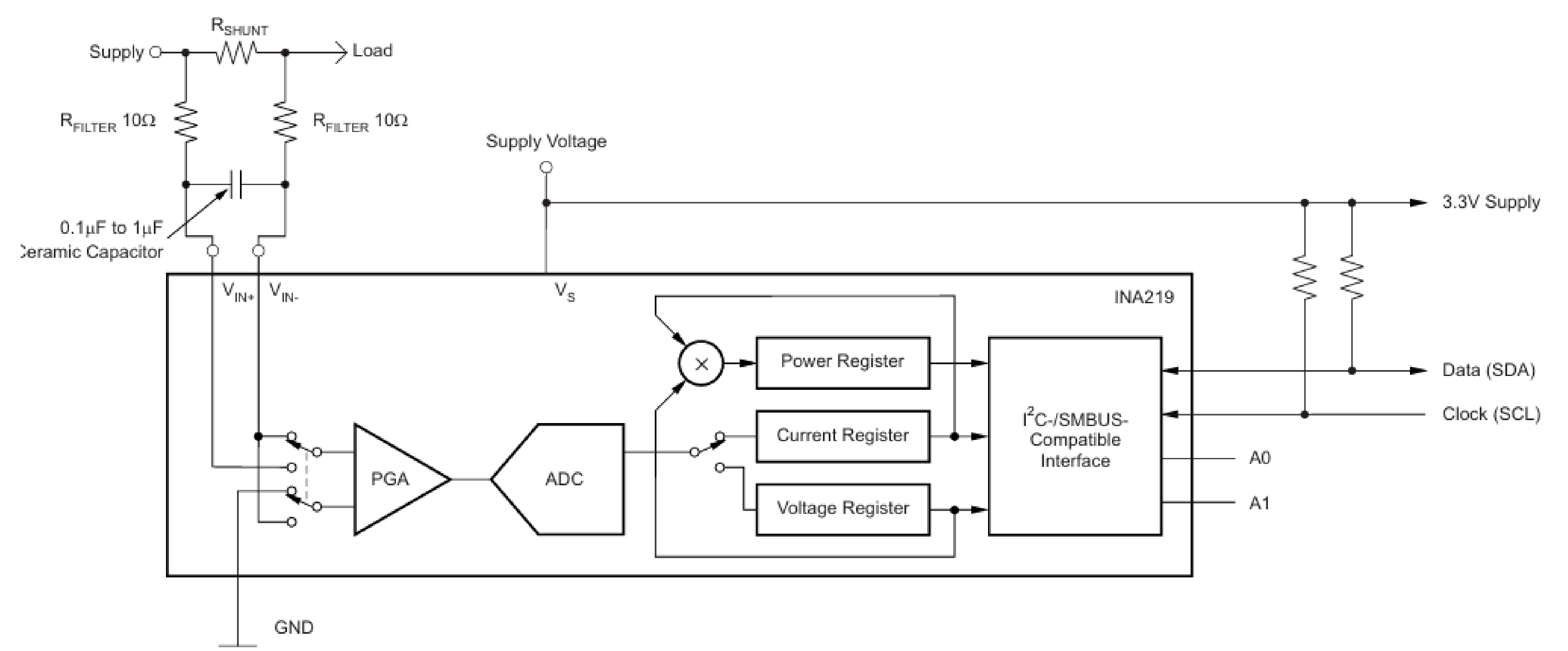


Fig.3 Customed Current Monitor Design

Main Components Overview

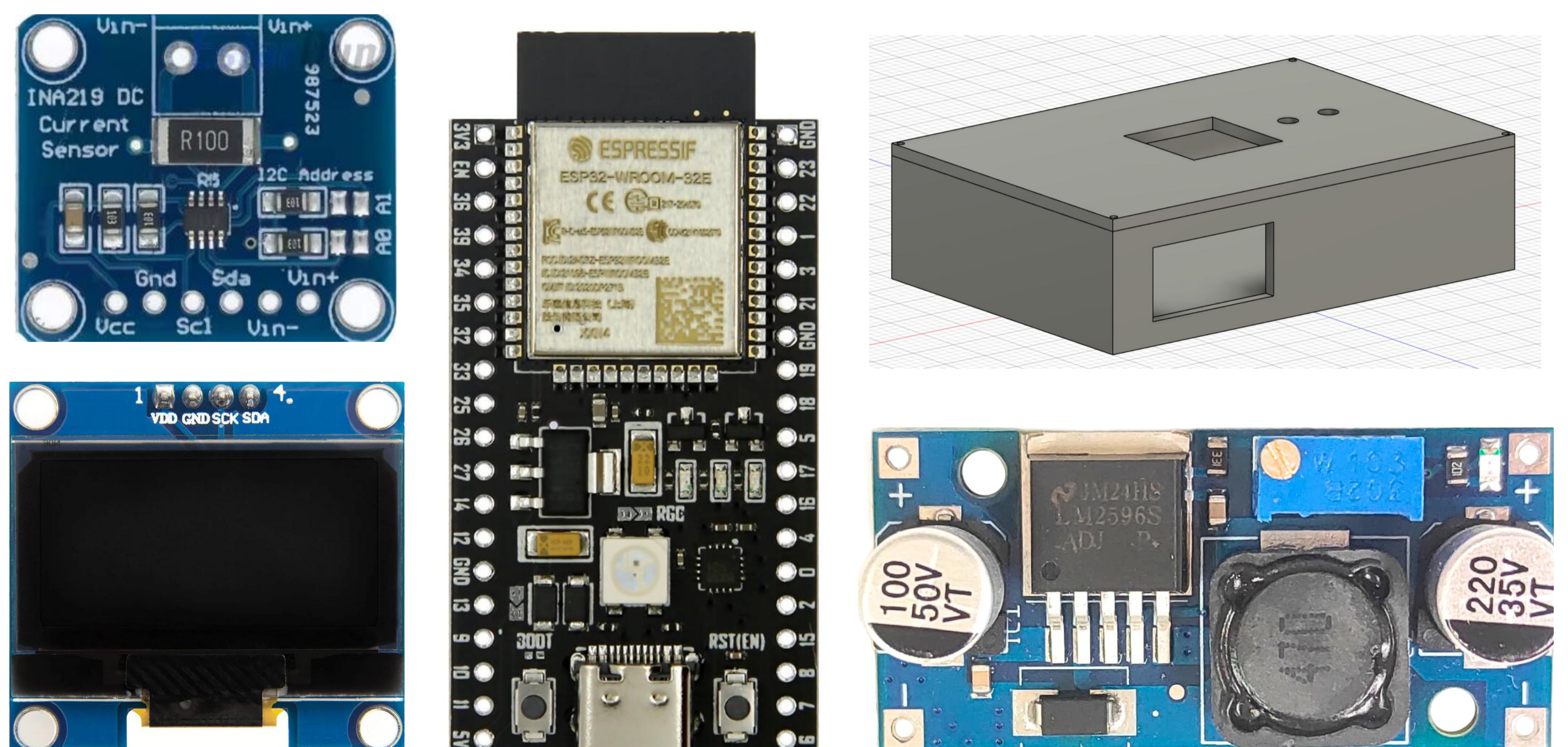


Fig.5 INA Based Current Monitor Module, OLED Display, ESP32 Microcontroller, 3D Printed Monitor Enclosure, LM2596 DC/DC Converter